LESSON PLAN

A/C ENGINEERING SYSTEM (17)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	02
Total No. of Lesson Plans	30	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Types of tanks, constructional details and arrangements in air frame.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Types of tanks, constructional details and arrangements in air frame.
5. Specific Objectives	17.1.1.1 Trainee will be able to understand the construction and requirements of typical aircraft fuel system and its components.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of fuel tank diagrams in IB; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of an aircraft fuel tank?
(b) Name two common locations for fuel tanks in an airframe.
(c) Why is the construction of fuel tanks critical for aircraft safety?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Internal vs External Tank Types • Bladder Tank Construction Details • Integral Tank Arrangement Features • Fuel Vent and Vapor Systems	L & Ds	PPT, IB	15 min
(b)	• Fuel Tank Heating Elements • Sump and Drain Valve Locations • Tank Pressure Relief Mechanisms • Anti-Corrosion Coating Standards	L & Ds	PPT, PDF	20 min
(c)	• Fuel Transfer Pump Arrangements • Scavenge System Operation • Tank Interconnect Valves • Emergency Fuel Dump Procedures	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between an integral tank and a bladder tank.
(b) Describe the function of the fuel tank sump in the airframe arrangement.
(c) How does the vapor vent system prevent tank over-pressurization?

12. Home Assignment and Follow Up	Prepare a brief note comparing turbine engine fuel requirements with piston engine fuels as covered in the next lesson plan.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Fuel, major classification of fuel based on fuel state, turbine engine fuels	Refer to the master syllabus reference list	LP 03

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Flight Lieutenant, P No. 12345	Senior Instructor, Wing Commander, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min